

Muhammad Huzaifa Aftab

Mechanical Engineering Undergraduate | Pakistan's Youngest Certified SOLIDWORKS Expert (CSWE) & SOLIDWORKS Champion
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PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate at UET Taxila and Pakistan's youngest Certified SOLIDWORKS Expert (CSWE) and Dassault Systèmes SOLIDWORKS Champion, with 20+ SOLIDWORKS and engineering certifications. Skilled in 3D CAD modeling, assembly design, GD&T, FEA, and CFD simulation, with hands-on manufacturing exposure from a five-department internship at Heavy Mechanical Complex (HMC) Taxila and freelance CAD/DFM consulting for international clients. Founder of MechMate Hub, a CAD/CAE mentoring platform supporting 50+ engineers worldwide, and a research contributor at MultiOmics leading literature review and technical writing for a manuscript on AI in Additive Manufacturing. Combines strong technical documentation and design-for-manufacturing skills with proven leadership across student engineering societies.

EDUCATION

B.Sc. Mechanical Engineering | University of Engineering & Technology Taxila *Aug 2023 – Present*

CGPA: 3.39 / 4.0 (up to 5th Semester)

HSSC — Pre-Engineering | Base Group of Colleges *Mar 2021 – Apr 2023*

Grade: 82.27%

TECHNICAL SKILLS

CAD & Design: 3D CAD Modeling & Assembly Design (SOLIDWORKS, Solid Edge), GD&T, Engineering Drawings, Reverse Engineering, DFM/DFMA

Simulation & Analysis: SOLIDWORKS Flow Simulation (CFD), Finite Element Analysis (FEA), Stress & Fatigue Analysis, Thermal Systems Analysis, Mesh Generation & Convergence Studies

Manufacturing: Machining, Foundry & Forge Processes, Production Planning, Quality Assurance/QC, Additive Manufacturing & 3D Printing

Programming & Tools: Arduino, PID Control Systems, HAP (HVAC Load Simulation)

Research & Documentation: Literature Review, Scientific & Technical Writing, Data Post-Processing, Engineering Reports

Project & Quality Management: Lean Six Sigma (Green Belt), Google Project Management, Primavera P6

Languages: English, German (Elementary)

RESEARCH EXPERIENCE

Research Intern | MultiOmics *2024 – Present • Remote*

- Serving as Group Leader for the review manuscript writing team, coordinating research tasks, aligning contributions across team members, and ensuring adherence to international academic publication standards under expert mentorship.
- Contributing to a review article on Artificial Intelligence Applications in Additive Manufacturing (AI-AM): Opportunities, Challenges, and Future Trends.
- Conducting literature review, critical analysis of research papers, citation verification, and technical writing to support manuscript development ahead of journal submission.

PROFESSIONAL EXPERIENCE

Summer Intern, Mechanical Engineering | Heavy Mechanical Complex (HMC) *Jul 2025 – Aug 2025 Taxila, Pakistan*

- Completed a six-week industrial internship with direct exposure to production planning, quality inspection, and large-scale mechanical systems.
- Observed and analyzed foundry, forging, CNC machining, and assembly operations, developing a practical understanding of manufacturing workflows applicable to physical domain modeling.
- Studied industrial fluid and thermal systems relevant to CFD boundary condition formulation.

Founder & Lead Mentor | MechMate Hub *Aug 2024 – Present • Remote*

- Established a technical mentoring platform supporting 50+ engineering students internationally in CAD/CAE and SOLIDWORKS certification preparation.
- Designed structured learning pathways and delivered SOLIDWORKS certification coaching (CSWA → CSWP → CSWE) through live coaching sessions.
- Built the MechMate Hub brand across LinkedIn, WhatsApp, and Fiverr, producing engineering education content on SOLIDWORKS, DFM/DFMA, and GD&T.

Freelance CAD Design Engineer | SolviqLab (Fiverr / Upwork) *Aug 2024 – Present • Remote*

- Deliver 2D-to-3D CAD modeling and precision design services for international clients through Fiverr and Upwork.
- Completed client projects including a screen lens/sealing assembly for G3 Inspection Solutions and a penetrometer-style gauge assembly.
- Manage client communication, technical scoping, and delivery independently while maintaining an active freelance profile and portfolio.

Chairperson & Lead Mentor | SOLIDWORKS Club, UET Taxila *Jul 2026 – Present Taxila*

- Lead the strategic growth and operations of the SOLIDWORKS Club, organizing webinars, workshops, and technical events on SOLIDWORKS and engineering design.
- Mentor students in 3D CAD modeling and SOLIDWORKS certification preparation (CSWA, CSWP, CSWE).
- Collaborate with faculty, industry professionals, and engineering communities to create learning opportunities.

PROJECTS

AAKRUTI Innovation Competition (Dassault Systèmes) – Team SERVENIX *2026*

- Competing in the AAKRUTI innovation competition with Team SERVENIX, designing a modular offloading insole system currently under development.
- Applying CAD modeling and product design methodology to a wearable, manufacture-ready device concept.

Arduino-Based Water Level PID Controller | Control Engineering CEP, UET Taxila

- Designed and implemented a PID-based control system for automated water level regulation as a Complex Engineering Problem (CEP).
- Tuned controller parameters and validated system response using Arduino hardware.

Cam-Driven Color Sorting Mechanism | Mechanics of Machines CEP, UET Taxila

- Designed a cam-actuated mechanism for automated color-based sorting, applying kinematic analysis and mechanism design principles.

HVAC Load & System Simulation, Commercial Office Building (Abu Dhabi) | Refrigeration & A/C Capstone, UET Taxila

- Performed HAP-based simulation of heating and cooling loads and system design for a commercial office building, with weekly progress reporting and full engineering documentation.

KEY ACHIEVEMENTS

- Youngest Certified SOLIDWORKS Expert (CSWE) and SOLIDWORKS Champion in Pakistan, globally recognized by Dassault Systèmes.
- Earned 20+ SOLIDWORKS and engineering certifications, demonstrating exceptional technical depth and commitment to excellence.
- Top 10 Performer, ASPIRE Pakistan Founders Edge Cohort III, recognized for leadership and innovation among a competitive national entrepreneurship cohort.
- Completed sustainability energy modeling training using the LEAP platform, applying computational tools to real-world energy and environmental systems.

CERTIFICATIONS

Certified SOLIDWORKS Expert (CSWE)	<i>Dassault Systèmes</i>
SOLIDWORKS Champion (Global Recognition)	<i>Dassault Systèmes</i>
Certified SOLIDWORKS Flow Simulation Professional (CSWP-Flow)	<i>Dassault Systèmes</i>
Certified SOLIDWORKS Simulation Professional (CSWP-S)	<i>Dassault Systèmes</i>
FEA Specialization: Stress Analysis, Heat Transfer in Fluids, Convergence & Mesh Independence	<i>Coursera</i>
Certified Istec Vibration Training Category 1+	<i>Istec International</i>
Google Project Management Professional Certificate (v2)	<i>Google / Coursera</i>
Primavera P6 — Project Scheduling	
Lean Six Sigma White, Yellow & Green Belt	

RESEARCH INTERESTS

Advanced Manufacturing Processes · Electrical Discharge Machining (EDM) · Advanced Aerospace Alloys · Sustainable Manufacturing Systems · Green Dielectric Fluids · Experimental Investigation · Process Optimization · Mechanical Design

PROFESSIONAL MEMBERSHIPS & LEADERSHIP

- SOLIDWORKS Champion, Dassault Systèmes Global Network
- Affiliate Member, Institution of Mechanical Engineers (IMechE)
- Student Member & Assistant Vice Chair (Media & Graphics), ASME UET Taxila Chapter
- Member, International Association of Engineers (IAENG)
- Media Secretary, UMEED-E-SUBH Welfare Society, UET Taxila (previously Team Lead, Co-Head, Executive)
- Graduate, Aspire Leaders Program — a Harvard-founded global leadership initiative delivered by Aspire Institute
- Entrepreneurship Trainee, ASPIRE Pakistan — Founders Edge Cohort III
- Facilitator (previously Intern), Nobel Learning PBC — Global Internship Program
- Student Volunteer, Fondi Inc — supporting a global online community for cross-cultural English conversation practice